

Technology Stack

Date	20/09/2026
Team ID	
Project Name	Study Buddy
Maximum Marks	4 Marks

Technology Stack

Layer	Technology	Purpose
Frontend	HTML, CSS, JavaScript / React	Responsive student interface
Backend	Node.js / Express	REST APIs and business logic
AI	Python AI service / LLM API	Explanations, summaries and quizzes
Database	MongoDB	Users, schedules, notes and progress
Security	JWT, bcrypt, HTTPS	Authentication and secure communication

System Characteristics

Characteristic	Implementation
Usability	Responsive and simple UI
Security	Authentication, hashing, HTTPS and access control
Performance	Optimized API/database queries
Scalability	Modular frontend, backend and AI services